- A suitable load M (eg; 100 g) is added to the mass hanger and the reading on the scale at which the pointer comes to rest is noted. This is the equilibrium position.
- The mass in the hanger is pulled downward and released so that the spring oscillates vertically on either side of the equilibrium position.
- When the pointer crosses the equilibrium position a stop clock is started and the time taken for 10 vertical oscillations is noted. Then the period of oscillation T is calculated.
- The experiment is repeated by adding masses in steps of 50 g to the mass hanger and period of oscillation at each time is calculated.
- For the masses M_1 and M_2 (with a difference of 50 g), their corresponding time periods are T_1 and T_2. Then the value $\frac{M_2 - M_1}{T_2^2 - T_1^2}$ is calculated and its average is found.
- Using the given formula the spring constant of the given spring is calculated.

OBSERVATIONS

Sl. No.	Mass M $\times 10^{-3}$ kg	Time taken for 10 oscillations (t) (s)			Period of oscillation $T = \frac{t}{10}$(s)	T^2 (s^2)	$\frac{M_2 - M_1}{T_2^2 - T_1^2}$ $\times 10^{-3} kg\ s^{-2}$
		Trial 1	Trial 2	Mean			
1	100						
2	150						
3	200						
4	250						
5	300						

Mean = ... kg s^{-2}

CALCULATION

Spring constant of the spring $k = 4\pi^2 \left(\frac{M_2 - M_1}{T_2^2 - T_1^2}\right)$

k = ... kg s^{-2}

RESULT

The spring constant of the given spring k is found to be = ... kg s^{-2}

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 290 02-01-2023 15:27:56